



Cambridge International AS & A Level

CANDIDATE
NAME
CENTRE
NUMBER

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CANDIDATE
NUMBER

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FURTHER MATHEMATICS

9231/33

Paper 3 Further Mechanics

May/June 2024

1 hour 30 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- Where a numerical value for the acceleration due to gravity (g) is needed, use 10 ms^{-2} .

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages. Any blank pages are indicated.

- (a) Show that $S = T - 3mg(1 + \cos \theta)$. [5]

[illegible]



[2]

[illegible]

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[7]

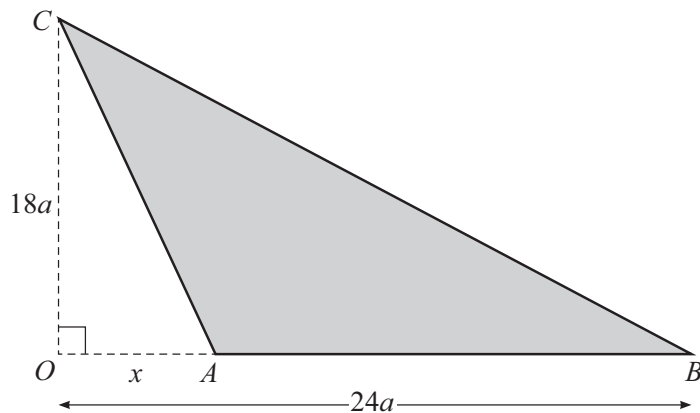
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[illegible]



[4]

[illegible]

- (a) Show that $\frac{7}{16}u^2 - 100u \sin \theta + 2500 = 0$. [3]

[illegible]



Find, in either order, the value of u and the value of $\sin \theta$. [5]

[illegible]

- (a) Find an expression for v in terms of t . [6]

[illegible]



(b) Find an expression for x in terms of t . [3]

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(c) Find the distance that the parachutist has fallen, since opening his parachute, when his speed is 15 m s^{-1} . [2]

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